

Math 261-001**Quiz 1, August 28**

Name	CSU ID #
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Be sure to read each question carefully. Answer all four problems. Each problem is worth the same amount of points. Write your final answer in the box provided. Only the answer written in the box will be graded. You may use the surrounding space for scratch work. For a plane equation, any nonzero scalar multiple of the same equation is equivalent.

1. Write $\vec{u} = (2, -1, 3)$ in the form $\vec{u} = a\mathbf{i} + b\mathbf{j} + c\mathbf{k}$. [10 points]

2. Let $\vec{u} = (1, 2, -1)$ and $\vec{v} = (3, -1, 2)$. Compute $\vec{u} \cdot \vec{v}$. [10 points]

3. Write a parametric equation $\vec{r}(t)$ for the line through $P = (1, -1, 2)$ and $Q = (3, 1, 1)$. [10 points]

4. Let $P = (1, 0, 1)$, $Q = (2, 1, 1)$, and $R = (1, 1, 2)$.

(a) Compute $\overrightarrow{PQ} \times \overrightarrow{PR}$.

[5 points]

- (b) Use the cross product from part (a) as a normal vector to write one equation of the plane through P , Q , and R . [5 points]